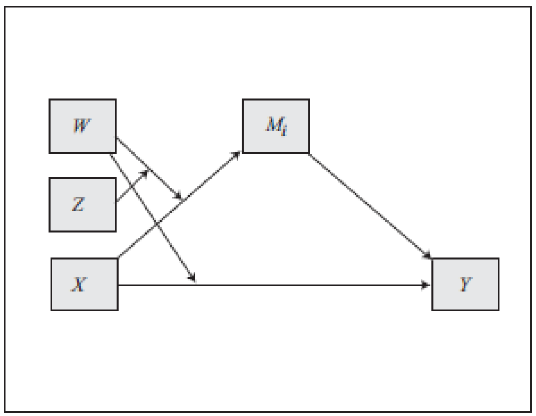
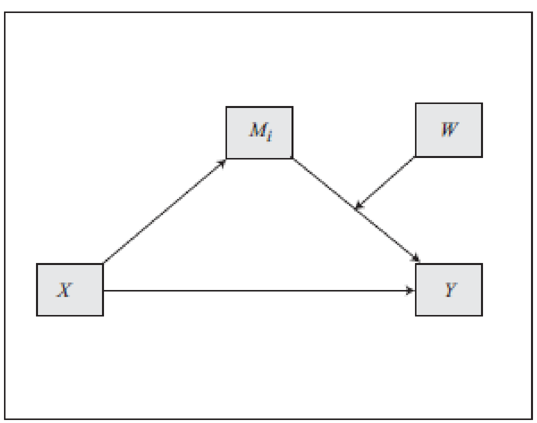
Constructing and editing models in PROCESS

Historically, PROCESS has operated by a model number system. The model numbers and the models those numbers represent can be found in the documentation. Choose the model number that corresponds to the model you would like to estimate.

Model 13



Model 14



Many of the preprogrammed numbered models you will find useful.

But what if the model you want to estimate does not correspond to any preprogrammed model represented by a model number?

Version 2: Too bad. Nothing you can do about it (unless you know some tricks).

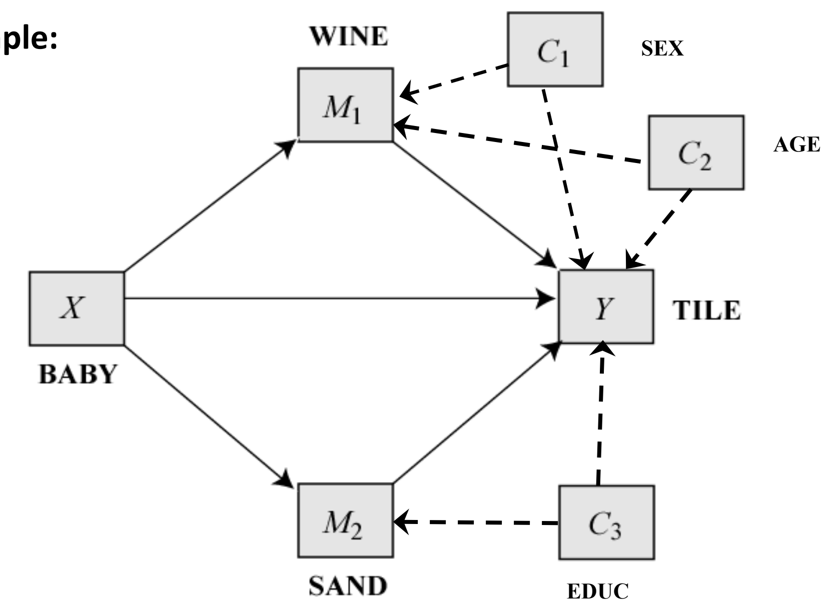
Version 3: Within certain constraints, you can create your own model from scratch, or edit an existing model number to make it correspond to the model you want to estimate.

Covariates are assignable to equations now

- Previous versions of PROCESS offer little flexibility in how covariates are assigned to equations. All covariates go in models of *Y* and mediator(s) *M*, or just *M*, or just *Y*. You can't split covariates up and assign them to different equations.

- With a new **cmatrix** option in version 3, covariates can now be assigned to different equations in whatever configuration you desire, rather than being forced to all be in the models of *M*s, *Y*, or both.

For example:




```
process y=tile/m=wine sand/x=baby/cov=sex age educ/model=4/
   cmatrix=1,1,0,0,0,1,1,1,1.
```

```
process(y="tile",m=c("wine", "sand"),x="baby",cov=c("sex", "age", "educ"),
model=4, cmatrix=c(1,1,0,0,0,1,1,1,1), data = fake).
```

Customizing the assignment of covariates

By default, all covariates are assigned to all equations, but this can be overridden with the **cmatrix** option. The assignment of covariates to equations is internally represented with the *C* matrix.

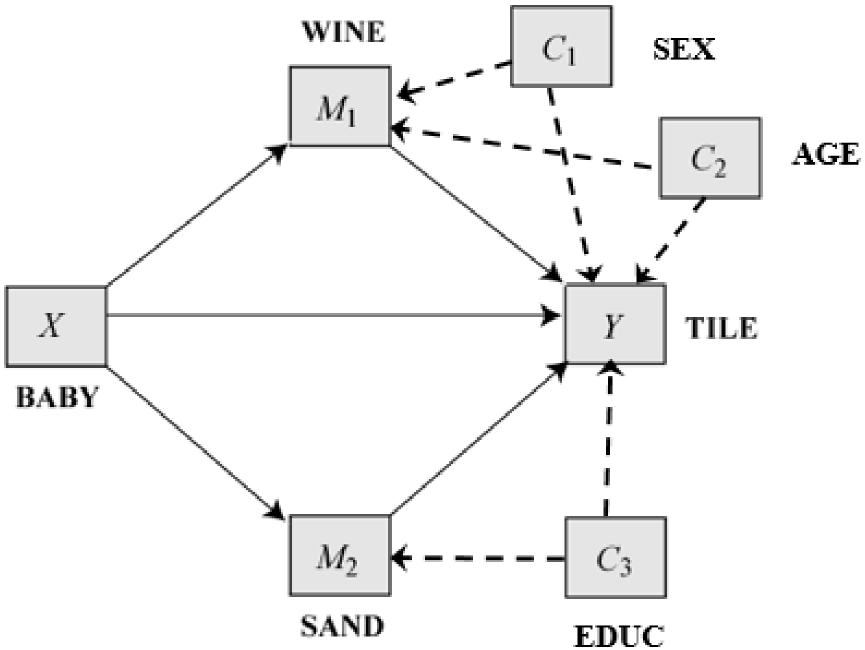
```
PROCESS y=yvar/x=xvar/m=med1 med2 ... medk/cov=cov1 cov2 … covk/model= ...
```

	cov1	cov2	…	covk
med1	0/1	0/1	…	0/1
med2	0/1	0/1	…	0/1
.	.	.	…	.
.	.	.	…	.
.	.	.	…	.
medk	0/1	0/1	…	0/1
yvar	0/1	0/1	…	0/1

C matrix

A one in the cell means the covariate in that column is to be included in the model of the variable in that row. A zero means the covariate in that column is to be excluded from the model of the variable in that row. By default, all entries in the *C* matrix are set to one.

Customizing the assignment of covariates



C matrix

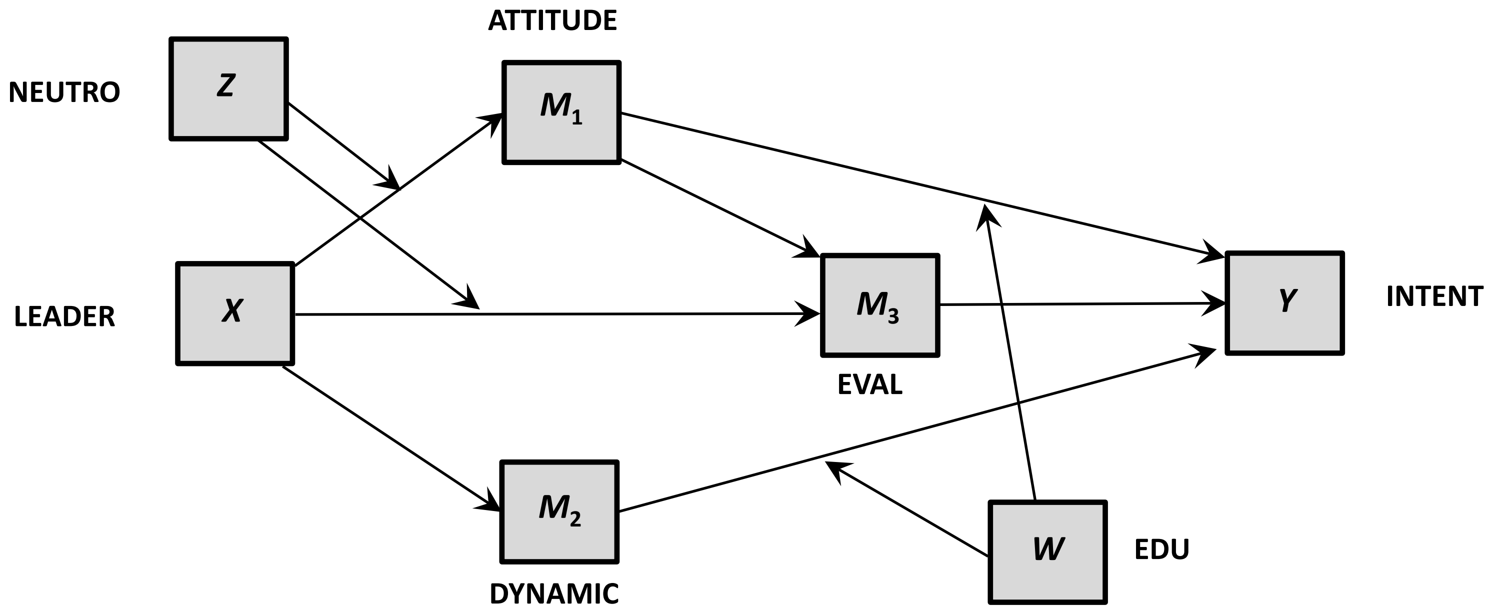
	sex	age	educ
wine	1	1	0
sand	0	0	1
tile	1	1	1

The **cmatrix** option does the assignment of covariates to equations. Read the matrix left to right, top to bottom, assigning the zeros and ones. Separate by commas in SPSS.

```
process y=tile/x=baby/m=wine sand/cov=sex age educ
/model=4 /cmatrix=1,1,0,0,0,1,1,1,1.
```

```
process(y="tile",m=c("wine", "sand"),x="baby",cov=c("sex", "age", "educ"),
model=4, cmatrix=c(1,1,0,0,0,1,1,1,1), data = fake).
```

A complex model



Such a complex model is not preprogrammed into PROCESS as a model number. In version 2, this model could not be estimated. In less than one hour, you will know how to program this model in PROCESS v3.

The *B* matrix

The *B* matrix is the heart of the representation of a model in PROCESS. It is a matrix of 0s and 1s specifying whether (1) or not (0) the variable in the column sends an effect to the variable in the row.

Variables sending effects (i.e., arrow points away)

Variables receiving effects (an arrow points at)

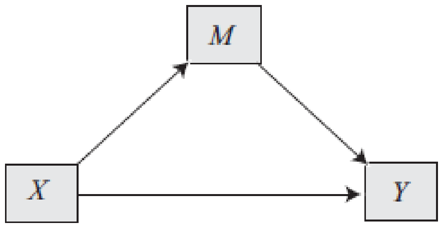
	X	M_1	M_2	$\ldots$	M_k
M_1	0/1	■	■	■	■
M_2	0/1	0/1	■	■	■
$\ldots$	0/1	0/1	0/1	■	■
M_k	0/1	0/1	0/1	0/1	■
Y	0/1	0/1	0/1	0/1	0/1

A model you program can contain one *X*, one *Y*, and up to 6 mediators. Certain cells are fixed to zero (the black squares above) to ensure the model is recursive (no feedback loops; PROCESS cannot estimate nonrecursive models).

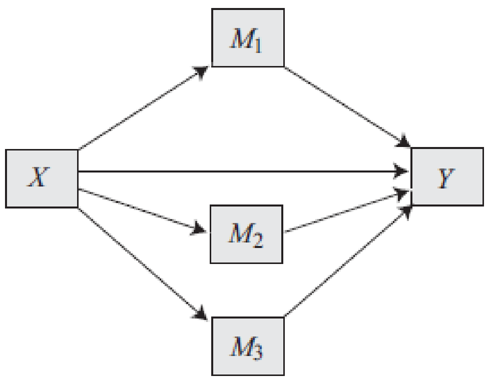
Examples

Model in conceptual form

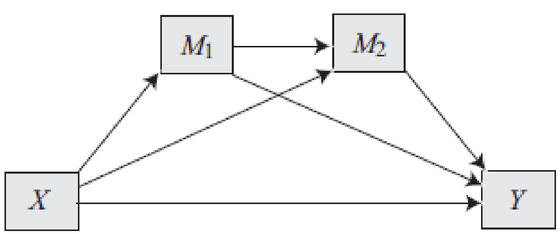
Model in *B* matrix form



	X	M
M	1	■
Y	1	1



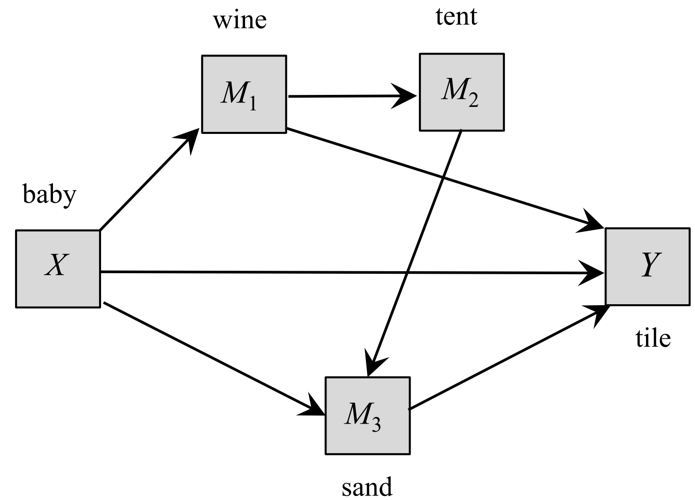
	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	0	0	■
Y	1	1	1	1

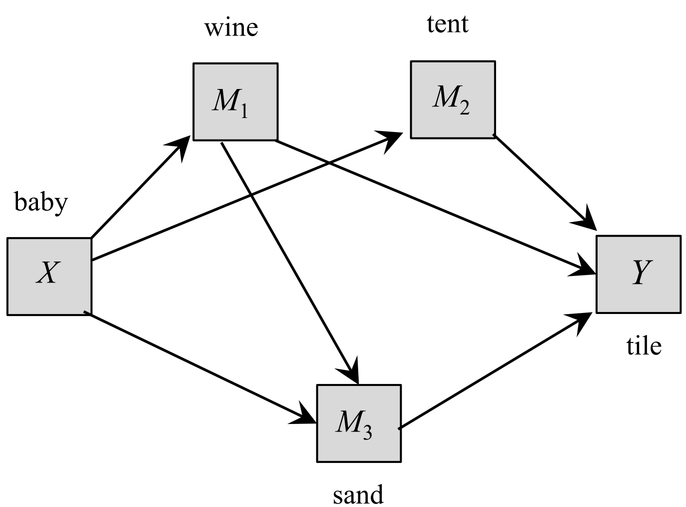


	X	M_1	M_2
M_1	1	■	■
M_2	1	1	■
Y	1	1	1

Examples

		X baby	M_1 wine	M_2 tent	M_3 sand
M_1	wine	1	■	■	■
M_2	tent	0	1	■	■
M_3	sand	1	0	1	■
Y	tile	1	1	0	1

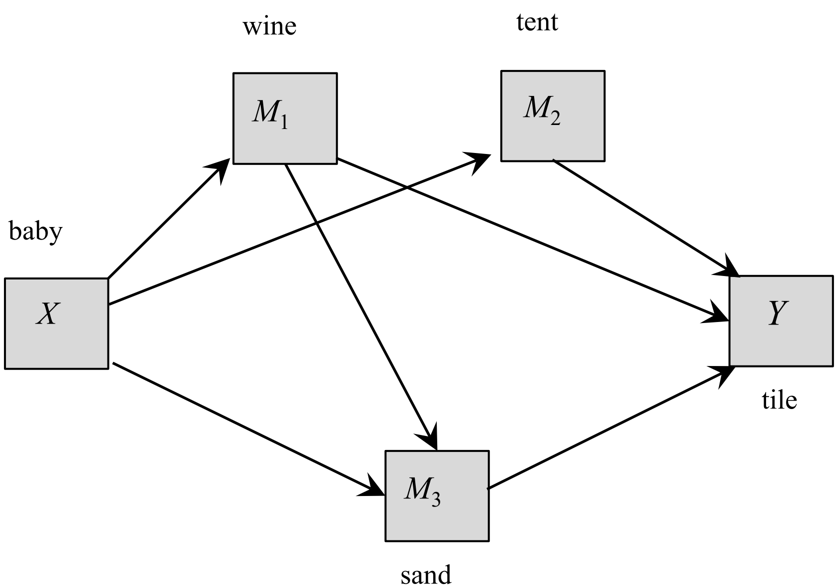




		X baby	M_1 wine	M_2 tent	M_3 sand
M_1	wine	1	■	■	■
M_2	tent	1	0	■	■
M_3	sand	1	1	0	■
Y	tile	0	1	1	1

Programming the *B* matrix

The mediation component of a model is programmed using the **bmatrix=** statement followed by a sequence of zeros and ones.



		X baby	M_1 wine	M_2 tent	M_3 sand
M_1	wine	1	■	■	■
M_2	tent	1	0	■	■
M_3	sand	1	1	0	■
Y	tile	0	1	1	1

Read the *B* matrix from left to right, top to bottom, skipping the black squares, and enter the zeros and ones in the sequence as they are encountered in the B matrix. Separate with commas in SPSS, but not in SAS, in R use c( ).

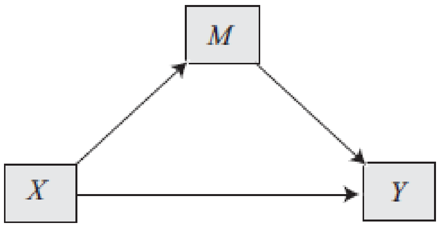
```
process y=tile/x=baby/m=wine tent sand/bmatrix=1,1,0,1,1,0,0,1,1,1.
```

```
process(data = fake, y="tile", x="baby", m=c("wine", "tent",
"sand"), bmatrix=c(1,1,0,1,1,0,0,1,1,1))
```

Examples

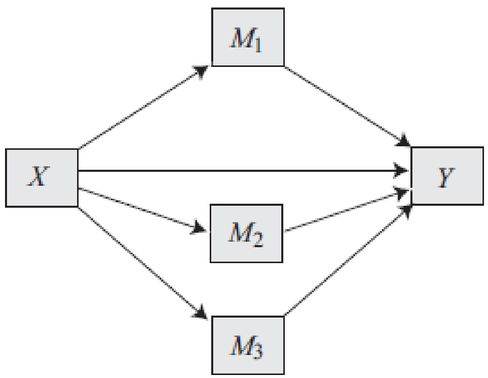
Model in conceptual form

Model in *B* matrix form



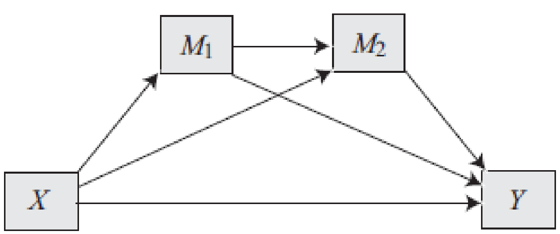
	X	M
M	1	■
Y	1	1

bmatrix=1,1,1



	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	0	0	■
Y	1	1	1	1

bmatrix=1,1,0,1,0,0,1,1,1,1



	X	M_1	M_2
M_1	1	■	■
M_2	1	1	■
Y	1	1	1

bmatrix=1,1,1,1,1,1

Moderation: The *W* and *Z* matrices

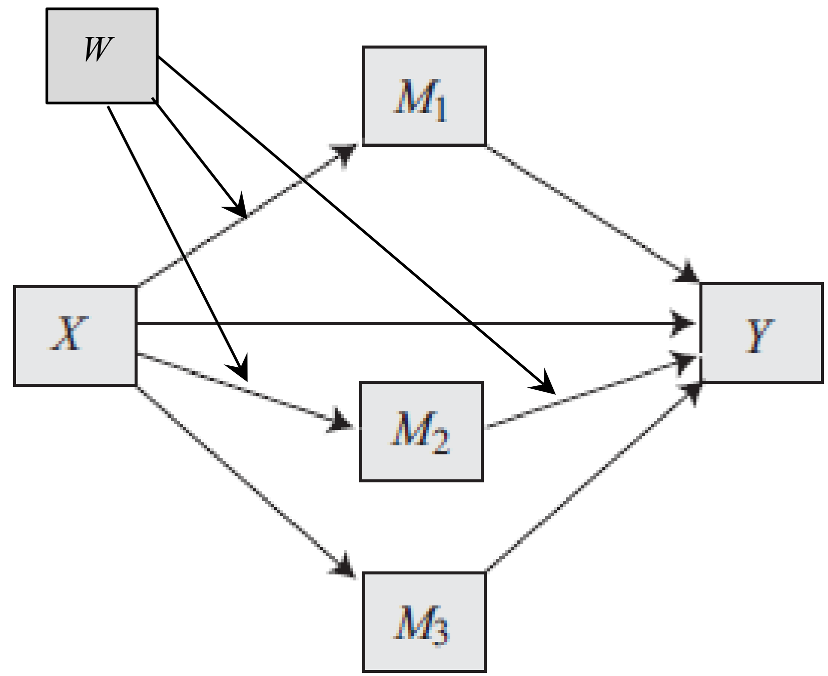
The *W* and *Z* matrices define which paths specified in the *B* matrix are moderated (1) and (0) not moderated. These matrices have the same form and size as the *B* matrix.

B matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	0	0	■
Y	1	1	1	1

W matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	0	0	0	■
Y	0	0	1	0



Note: A path fixed at zero in the *B* matrix cannot be moderated.

Moderation: The *W* and *Z* matrices

A second moderator *Z* can be specified. But *Z* can only be used if *W* has already been used. That is, if your model has only one moderator, it must be called *W*.

B matrix

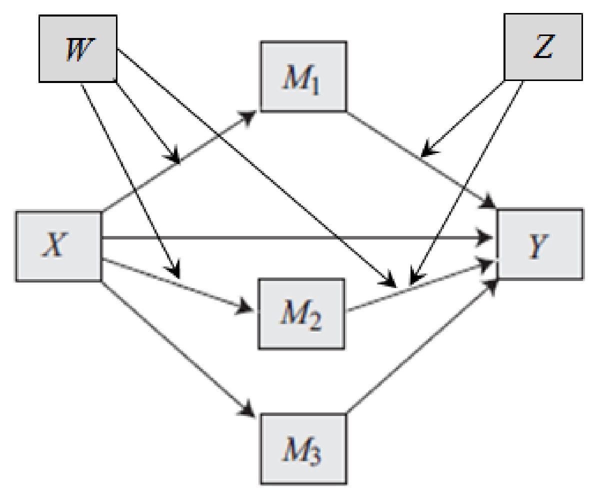
	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	0	0	■
Y	1	1	1	1

W matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	0	0	0	■
Y	0	0	1	0

Z matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	1	1	0



Programming the *W* and *Z* matrices

B matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	0	0	■
Y	1	1	1	1

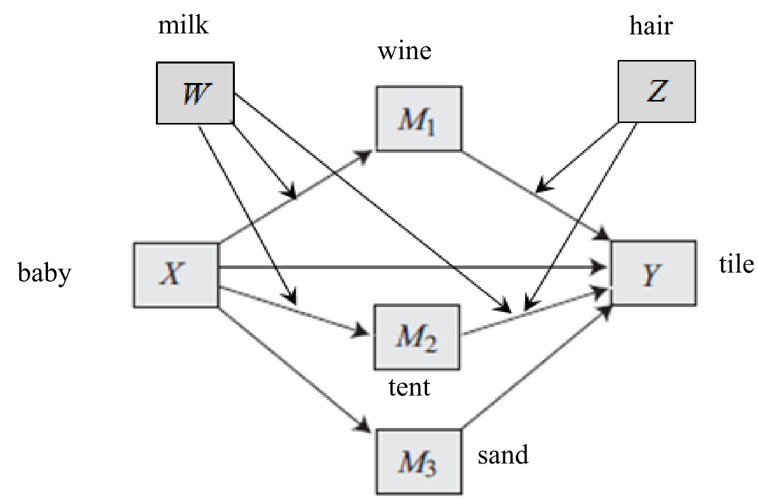
W matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	0	0	0	■
Y	0	0	1	0

Z matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	1	1	0

The *W* and *Z* matrices are programmed with the **wmatrix=** and **zmatrix=** statements. Use the same procedure as for the *B* matrix.



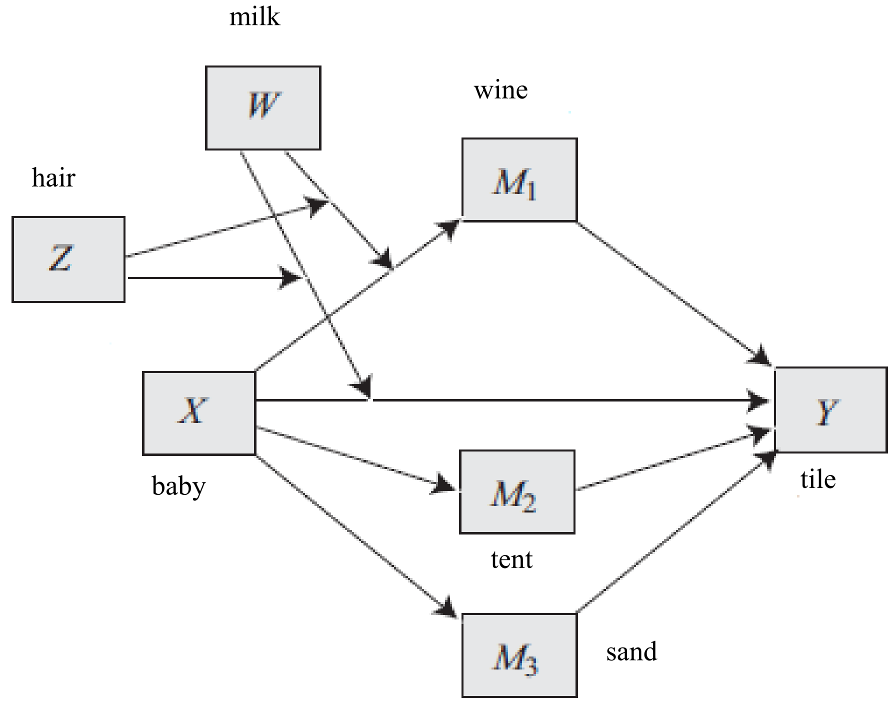

```
process y=tile/x=baby/m=wine tent sand/w=milk/z=hair
   /bmatrix=1,1,0,1,0,0,1,1,1,1/wmatrix=1,1,0,0,0,0,0,0,1,0
   /zmatrix=0,0,0,0,0,0,0,1,1,0.
```

Moderated moderation: The *WZ* matrix

Moderated moderation, also known as three-way interaction, is held in the *WZ* matrix. *W* is the primary moderator, and *Z* is the secondary moderator. Program using the **wzmatrix=** statement, using the same 0/1 system.

WZ matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	1	0	0	0




```
process y=tile/x=baby/m=wine tent sand/w=milk/z=hair/
   bmatrix=1,1,0,1,0,0,1,1,1,1/wzmatrix=1,0,0,0,0,0,1,0,0,0.
```

Note: if you don’t use **wmatrix** or **zmatrix**, this implies all cells are zero

Putting it all together

This system allows for the construction of some very complex models.

B matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	0	0	■
Y	1	1	1	1

W matrix

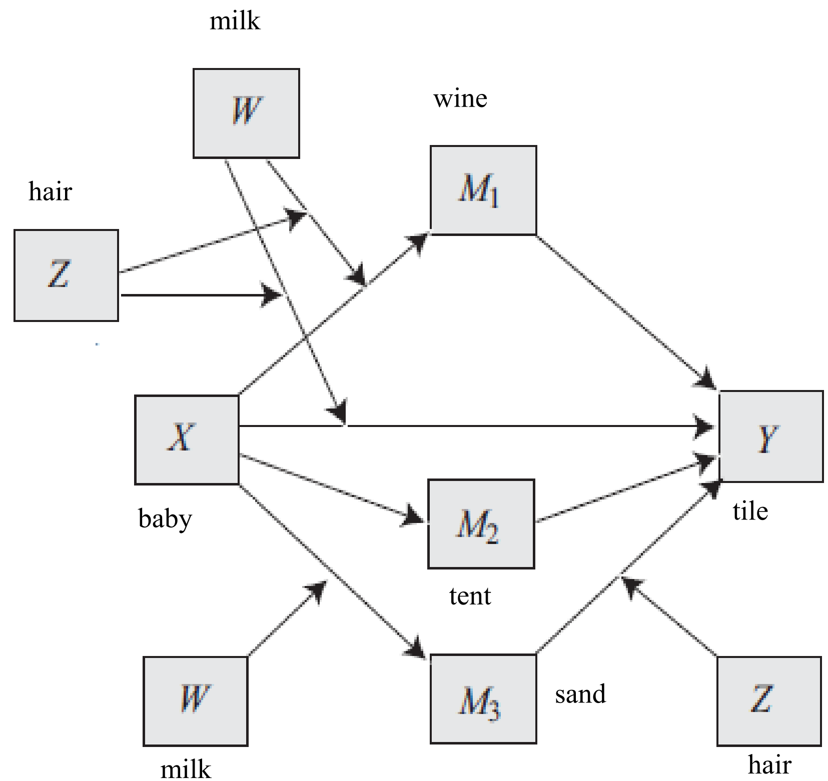
	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	1	0	0	■
Y	0	0	0	0

Z matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	0	0	1

WZ matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	1	0	0	0

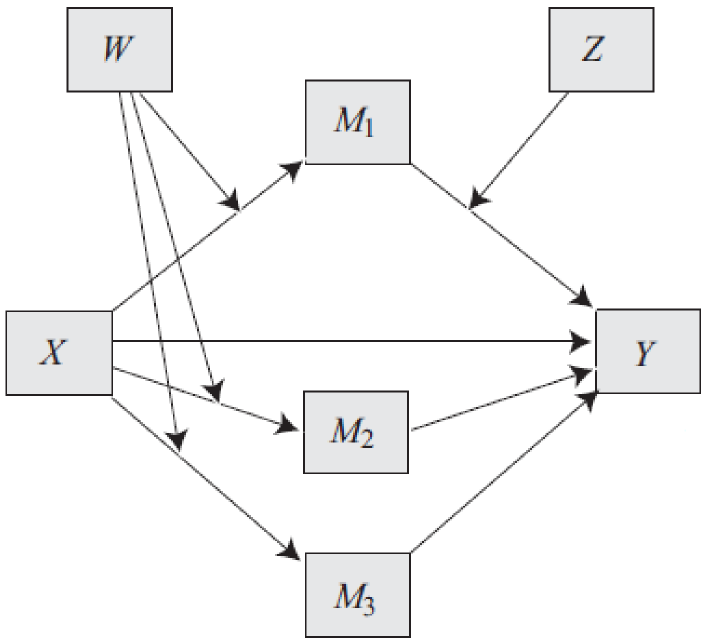



```
process y=tile/x=baby/m=wine tent sand/w=milk/z=hair
  /bmatrix=1,1,0,1,0,0,1,1,1,1/wmatrix=0,0,0,1,0,0,0,0,0,0
  /zmatrix=0,0,0,0,0,0,0,0,0,1/wzmatrix=1,0,0,0,0,0,1,0,0,0.
```

Editing a numbered model

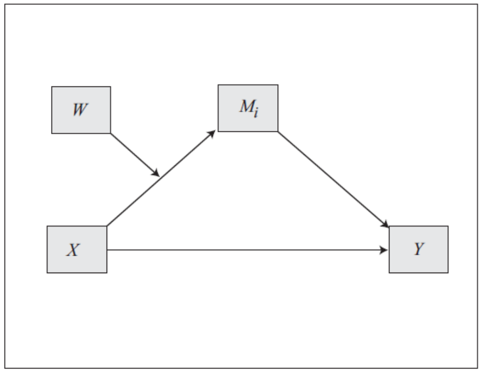
Many preprogrammed numbered models are likely to be close to the model you want to estimate. You can edit a modeled number, adding a desired interaction, or removing one you don't want. This is done by reprogramming the *W*, *Z*, and/or *WZ* matrices.

Example



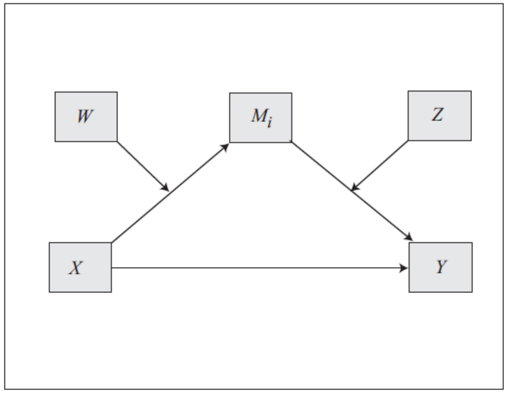
This is like model 7, except model 7 doesn't include moderation of any *M* to *Y* paths.

Model 7



This is like model 21, except model 21 would includes moderation by *Z* of the M_2 and M_3 to *Y* paths as well as the M_1 to *Y* path.

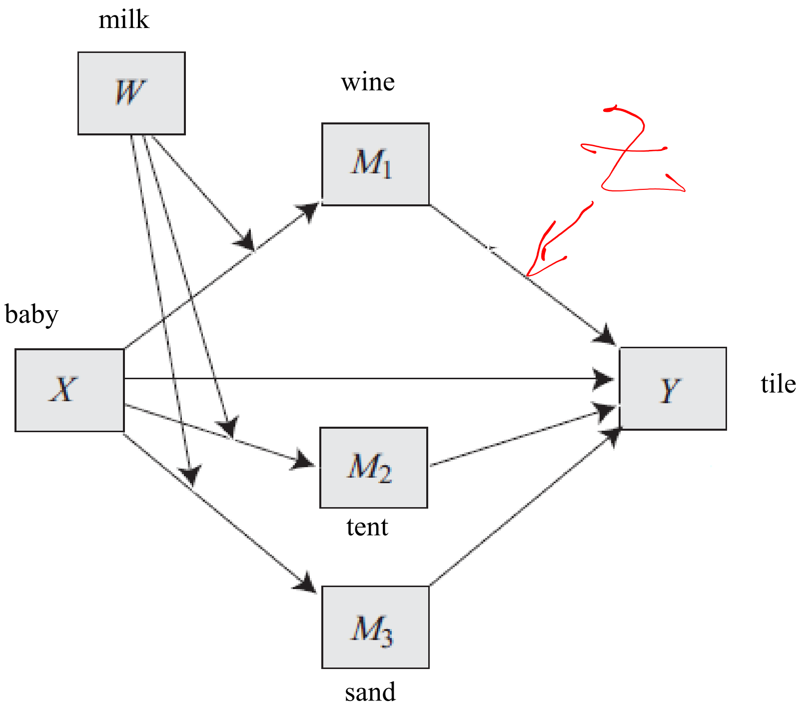
Model 21



NOTE: When using a model number, the *B* matrix cannot be edited.

Option 1: Edit model 7

In model 7, the *Z* matrix is all zeros because there is no *Z* in model 7. So program the *Z* matrix to include the moderation of the desired path by *Z*.



Preprogrammed Z matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	0	0	0

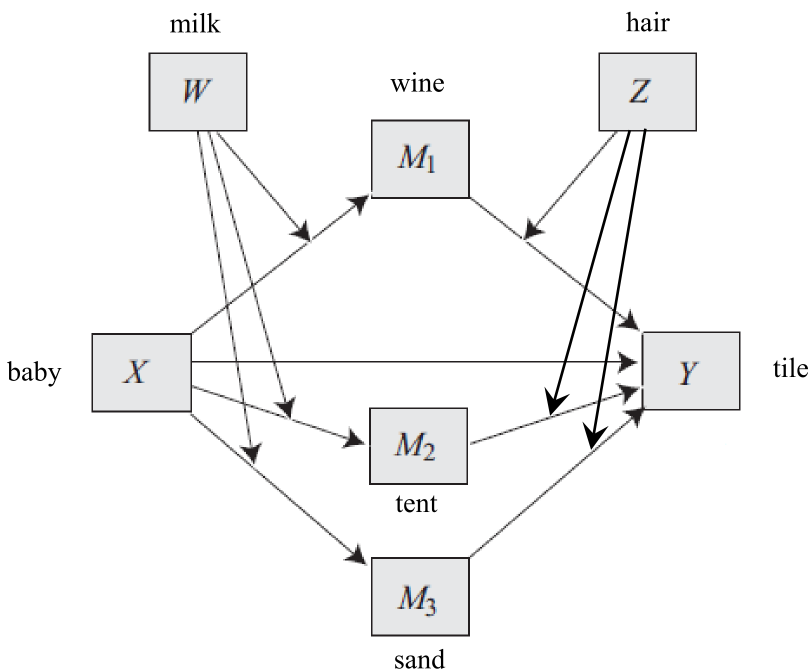
Desired *Z* matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	1	0	0

```
process y=tile/x=baby/m=wine tent sand/w=milk/z=hair/model=7/
    zmatrix=0,0,0,0,0,0,0,1,0,0.
```

Option 2: Edit model 21

In model 21, the Z matrix contains ones in certain cells that allow the M_2 and M_3 paths to Y to be moderated by Z. We can reprogram the Z matrix, turning the offensives ones into zeros.



Preprogrammed Z matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	1	1	1

Desired Z matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	1	0	0

```
process y=tile/x=baby/m=wine tent sand/w=milk/z=hair/model=21/
    zmatrix=0,0,0,0,0,0,0,1,0,0.
```

The MATRICES statement

If you want to check to make sure you have programmed the matrices correctly, or you want to see what the matrices of a preprogrammed model look like, add **matrices=1** to a PROCESS command.

```
process y=tile/x=baby/m=wine tent
  sand/w=milk/z=hair/model=7/
  zmatrix=0,0,0,0,0,0,0,1,0,0
  /matrices=1.
```

Matrices that don't appear in the output have zeros in all cells. If your model includes covariates, the *C* matrix will appear here too.

```
********** MODEL DEFINITION MATRICES **********

BMATRIX: Paths freely estimated (1) and fixed to
zero (0):
     baby wine tent sand
wine  1
tent  1    0
sand  1    0    0
tile  1    1    1    1

WMATRIX: Paths moderated (1) and not moderated
(0) by W:
     baby wine tent sand
wine  1
tent  1    0
sand  1    0    0
tile  0    0    0    0

ZMATRIX: Paths moderated (1) and not moderated
(0) by Z:
     baby wine tent sand
wine  0
tent  0    0
sand  0    0    0
tile  0    1    0    0
```

Exercise #1

What model does this set of matrices represent?
Draw a path diagram with all variables labeled

B Matrix

	X	M_1	M_2
M_1	1	■	■
M_2	1	1	■
Y	0	1	1

W Matrix

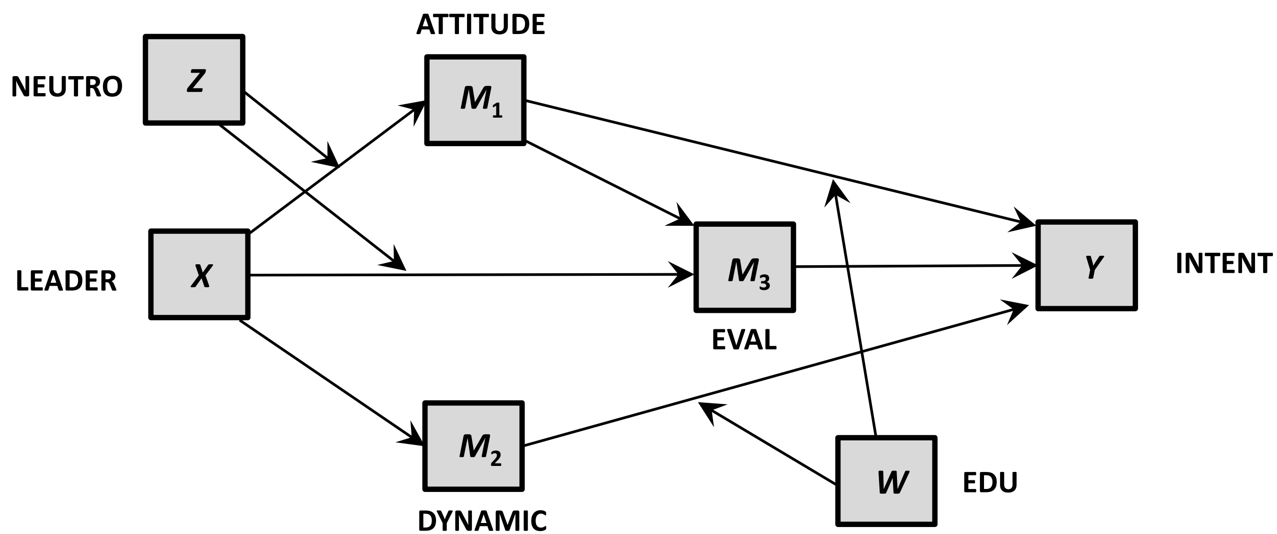
	X	M_1	M_2
M_1	1	■	■
M_2	0	1	■
Y	0	0	1

C Matrix

	U_1	U_2	U_3
M_1	1	0	0
M_2	0	1	1
Y	1	1	0

Exercise #2

Write the PROCESS command that estimates this model:



Exercise #1

What model does this set of matrices represent?

B Matrix

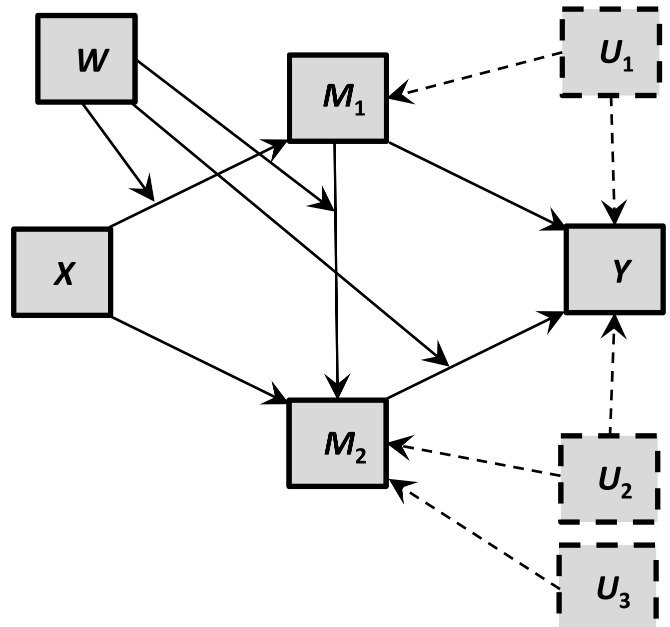
	X	M_1	M_2
M_1	1	■	■
M_2	1	1	■
Y	0	1	1

W Matrix

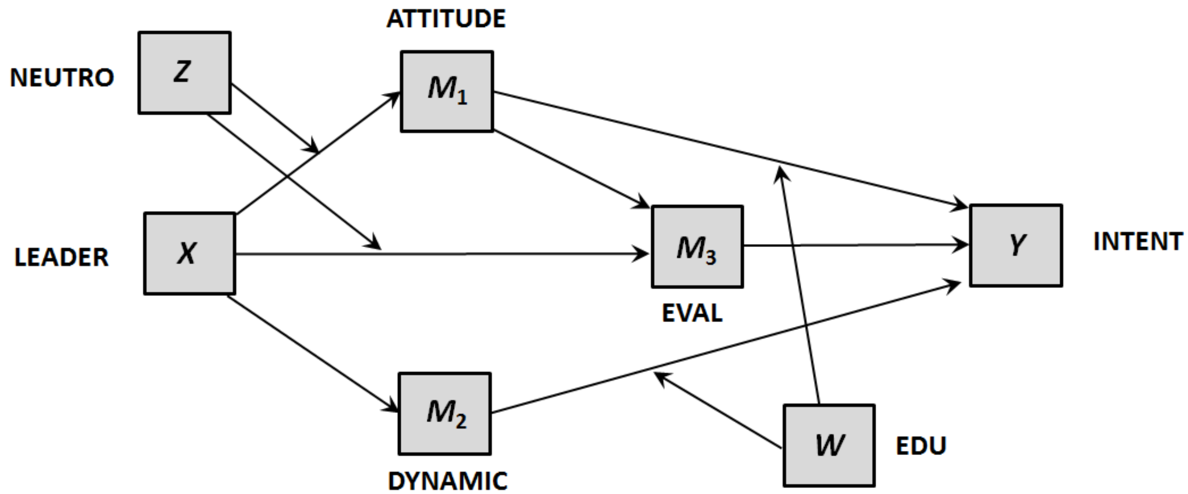
	X	M_1	M_2
M_1	1	■	■
M_2	0	1	■
Y	0	0	1

C Matrix

	U_1	U_2	U_3
M_1	1	0	0
M_2	0	1	1
Y	1	1	0



Exercise #2



B matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	1	0	■	■
M_3	1	1	0	■
Y	0	1	1	1

W matrix

	X	M_1	M_2	M_3
M_1	0	■	■	■
M_2	0	0	■	■
M_3	0	0	0	■
Y	0	1	1	0

Z matrix

	X	M_1	M_2	M_3
M_1	1	■	■	■
M_2	0	0	■	■
M_3	1	0	0	■
Y	0	0	0	0

```
process y=intent/x=leader/m=attitude dynamic eval /w=edu /z=neutron
/bmatrix=1,1,0,1,1,0,0,1,1,1 /wmatrix=0,0,0,0,0,0,0,1,1,0
/zmatrix=1,0,0,1,0,0,0,0,0,0.
```